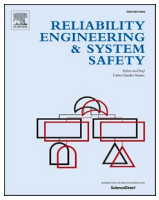
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A dynamic reliability assessment method for multi-state manufacturing system by merging imprecise observational information

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ABSTRACT

Accurate reliability assessment of advanced manufacturing systems is essential for ensuring production efficiency, reducing downtime, and enabling intelligent maintenance strategies. In practical industrial environments, state observations obtained from sensors or expert evaluations are often imprecise. Effectively utilizing this uncertain information can substantially improve the precision of reliability evaluations. Conventional methodologies often encounter limitations in addressing this challenge, as manufacturing systems are generally characterized by networked production line configurations rather than traditional serial or parallel structures. Moreover, the effective integration of imprecise observational data is essential for the continuous updating of system reliability. This study introduces a novel approach for dynamic reliability evaluation of a multi-state manufacturing system (MSMS), incorporating both rework mechanisms and buffer elements to enhance the accuracy and applicability of system reliability assessments. The MSMS model can effectively depict the gradual degradation processes and diverse performance levels of manufacturing systems, allowing for a more realistic and detailed representation of system behavior over time compared to traditional binary-state models. This study employs the multistate flow network (MFN) model to construct the MSMS reliability assessment framework from a network structure perspective. Dynamic Bayesian networks (DBNs) are developed to update the reliability function of an individual MSMS by incorporating evidential observational data. An illustrative case study on the reliability update of an aluminum alloy wheel production line is presented to demonstrate the proposed methodology. The case study results further confirm the effectiveness of the approach.

1. Introduction

In modern manufacturing systems, machines functioning as manufacturing units are interconnected within networked production lines, with each unit executing a specific process and collectively determining overall system output. Although reliability is a critical metric for guiding production planning and maintenance, traditional assessment methods, typically based on serial or parallel models, are insufficient for accurately evaluating advanced manufacturing systems composed of complex networks of interconnected machines. Therefore, it is highly valuable to study how to establish a reliability assessment model for production-line-type manufacturing systems from the perspective of network structures. The majority of existing research on manufacturing system reliability assessment emphasizes the development of complex modeling frameworks to more accurately reflect real-world conditions and thereby enhance evaluation precision. This study posits that incorporating information derived from real-time state observational data to update system reliability constitutes an equally effective strategy for improving the accuracy of reliability assessments. Since the performance degradation of manufacturing networks is stochastic, to improve the accuracy of reliability assessment, real-time state observational data can be used to enable dynamic evaluation of manufacturing systems. This step shifts the focus of reliability

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Nomenclature

$\boldsymbol{S}$	Set of the states of manufacturing units
$\boldsymbol{B}$	Set of the buffers
$\boldsymbol{P}$	Set of the manufacturing units' yield rates
$\boldsymbol{T}$	Set of the tasks
m	Manufacturing unit
b	Buffer
p	Yield rate of manufacturing units
t	Task
s_{ij}	The jth state of the manufacturing units m_i
c_{ij}	The output capacity of manufacturing unit m_i when operating in state s_{ij}
$x_i^{\min}$	The minimal capacity of manufacturing unit m_i
$rw_{f_i}^{m_i}$	The rework of manufacturing units m_i
Rw_{Oi}	The qualified products output by rework $rw_{f_i}^{m_i}$
I	Input quantity of raw materials
O	Output quantity of finished products
Pt	Path in the manufacturing system
MPt	Minimal path in the manufacturing system
$Pt*$	Feasible path in the manufacturing system
r^*	Probability of feasible path
X	Node in the Bayesian network
$\mathrm{Pa}(X)$	The parent node of X
Ω	Node set of the Bayesian network
$\Pr\{\cdot\}$	Joint probability distribution
$R\{\cdot\}$	Reliability function
Θ	Framework of discernment
$m(\cdot)$	Mass function
$Bel(\cdot)$	Belief function
$Pl(\cdot)$	Plausibility function
$\mathrm{pl}(\cdot)$	Contour function
$\widetilde{\mathbf{o}}(t)$	Set of evidential observation at time instant t
$\mathbf{p}^{C_l}(t)$	Set of probability distribution of component nodes at time instant t
P^{m_i}	State probability distribution of manufacturing units m_i
P^{m_i}	Updated state probability distributions of manufacturing units
$\mathbf{m}^{C_l}(t)$	Set of mass function of component nodes at time instant t
$\mathbf{m}_{updated}^{c_l}(t)$	Set of updated mass function of component nodes at time instant t
MSMS	Multi-state manufacturing system
MSS	Multi-state system
MFN	Multistate flow network
DBN	Dynamic Bayesian network
BN	Bayesian network
WIP	Work-in-process
MP	Minimal path
CPT	Conditional probability table
FoD	Framework of discernment
DRC	Dempster rule of combination
TM	Transition matric

assessment from a generic class of systems to a specific system in operation. In engineering practice, state observational data of manufacturing systems mainly comes from individual manufacturing units. These observations may originate from different hierarchical levels of the units and are often imprecise due to sensor noise or expert uncertainty. Although several methods, such as hybrid Bayesian frameworks [1] and dynamic epistemic reliability models [2], have been developed for reliability assessment under imprecise observations, they are often designed for relatively simple system configurations and fail to fully address the complexity of advanced manufacturing systems. In such systems, manufacturing units are interconnected through rework flows, buffering mechanisms, and various operational couplings, substantially increasing the complexity of propagating unit-level uncertainties to the system level. Consequently, a specialized methodology is required that concurrently captures the networked operational architecture of manufacturing systems and integrates the inherent uncertainty of real-world observational data, thereby facilitating accurate and dynamic assessment of system-level reliability.

To address this challenge, a hybrid methodology is proposed that combines the multistate flow network (MFN) model with dynamic Bayesian networks (DBNs), supported by evidence theory. The MFN framework is utilized to represent the interconnected structure and flow-oriented operational characteristics of multi-state manufacturing systems, while the DBN is employed to dynamically update the reliability of individual units by incorporating imprecise observational data, quantitatively interpreted through evidence theory. Specifically, in manufacturing systems, the manufacturing units are primarily various types of machine tools that do not fail abruptly but go through intermediate states of partial performance degradation. Hence, the manufacturing units are modeled as traditional hierarchical multi-state systems (MSSs), and the manufacturing system is regarded as a multi-state manufacturing system (MSMS) composed of multi-state manufacturing units. An MSS refers to a system whose components or the system as a whole can operate at finite distinct performance levels. An MSMS is a manufacturing system modeled under the multi-state system framework, in which the manufacturing unit and the whole manufacturing system can function at multiple capacity levels. Therefore, in this study, the MSS refers to the unit-level system, while the MSMS denotes the network-level system. The reliability of the MSMS is fundamentally determined by the reliability of each MSS and the nature of its interconnections. Accordingly, the MFN model is employed to construct the MSMS reliability assessment framework, as it effectively captures the flow-oriented structural characteristics of manufacturing systems and accommodates the multi-state performance behavior of individual units. Meanwhile, since the uncertainty in state observation data primarily stems from the difficulty in accurately determining the states of manufacturing units, the evidence theory is employed to quantify and propagate this epistemic uncertainty. The evidence theory is suitable for this task, as it uses mass functions to effectively represent the probabilities of different element sets. Finally, the DBNs are specifically designed to update the reliability function of an individual multi-state manufacturing unit by merging evidential observational data and further update the overall manufacturing system reliability through the MFN model built before.

Building on the above concepts and methods, this paper aims to develop a novel dynamic reliability assessment method for production line-based MSMS. The main objectives are to accurately model the multi-state characteristics of manufacturing systems, considering real operational features such as rework processes and buffer mechanisms, and to enable dynamic reliability updating by integrating both precise and imprecise observational data from manufacturing units at different hierarchical levels. The innovation of this study lies in the establishment of a dynamic reliability assessment framework for MSMSs, integrating both network-based system modeling and imprecise observational data. In contrast to traditional methods that assume static reliability and require precise input parameters, this research introduces a hybrid approach that combines the MFN model to characterize the structural and operational features of production lines with the DBN to facilitate continuous reliability updates over time. The primary contribution of this study is the development of a novel methodology for dynamic

reliability evaluation of manufacturing systems using imprecise state observation data, thereby addressing the limitations of conventional static approaches that rely solely on historical information.

The remainder of this article is structured as follows: Section 2 is the literature review. Section 3 introduces the MSMSs in question and the corresponding MFN model. The dynamic reliability assessment for each individual manufacturing unit by imprecise observation data is presented in Section 4. The proposed reliability updating method is described in detail in Section 5. In Section 6, a case study involving the production line of aluminum alloy wheels is presented as an illustrative example. Finally, Section 7 concludes the paper with a summary and discussion of future research directions.

2. Literature review

In recent years, an increasing number of researchers have adopted a network-based perspective to analyze production lines [3,4]. By representing manufacturing systems as structured networks, comprehensive analyses can be systematically performed within this framework, facilitating a deeper understanding of system behavior and performance characteristics [5]. Lee and Garcia-Diaz applied a flow-network method to optimize machine grouping in a manufacturing system, leveraging functional similarity as a key performance metric [3]. Expanding on this concept, an extended modeling approach was introduced, treating manufacturing systems as multi-state manufacturing networks to account for the stochastic capacities of individual equipment [6]. Building upon these network-based models, researchers have advanced the exploration of key dimensions such as reliability evaluation, health monitoring, and service life prediction, thereby contributing to improved efficiency and long-term sustainability of manufacturing systems.

Lin and Yuan [6] investigated the reliability assessment of multi-state flow networks. By incorporating graph theory concepts, they evaluated network reliability by identifying the minimum path, establishing a foundation for future reliability assessments of manufacturing systems using network models. Building upon this foundation, Lin [7] later introduced a system reliability assessment method based on the stochastic flow network model. By identifying the minimum cut sets, this approach enables the evaluation of system reliability under varying task requirements, offering a more dynamic and adaptable framework for reliability analysis. Further advancing these developments, Lin and Chang [8] applied this reliability assessment methodology to manufacturing networks, integrating rework actions within the production line to enhance system performance evaluation. Furthermore, various factors, including rework [9], joint buffers [10], multiple production lines [11], and advanced intelligent algorithms [12,13] were considered in the process of refining the model, improving its accuracy and applicability in complex manufacturing environments..

He et al. [14] introduced an extended MFN method to evaluate the mission reliability of the manufacturing system, incorporating the impact of product quality states on system reliability. Han et al. [15] further explored the remaining useful life prediction and the optimization strategy for predictive maintenance in manufacturing systems. Their study proposed an approach for calculating average maintenance costs based on mission reliability evaluation within manufacturing networks. Liao et al. [16] proposed a predictive maintenance strategy for manufacturing systems based on operational risk, incorporating factors such as product quality degradation and delays in production tasks. Their research highlighted the essential role of mission reliability assessment in evaluating overall system performance and informing the development of optimized maintenance plans.

Most existing research has focused on developing manufacturing system models that accurately represent real-world conditions and applying these models for subsequent analysis. However, comparatively limited attention has been directed toward constructing dynamic reliability assessment models specifically designed for manufacturing systems. In prior studies, the state probability distributions of individual units within manufacturing systems are often assumed to remain static, with reliability evaluated at discrete points in time. This paper seeks to explore a dynamic reliability updating approach for production line-based manufacturing systems from the perspective of continuous evaluation. Furthermore, when a manufacturing system exhibits multi-state characteristics, a key research challenge is effectively capturing these multi-state features and determining a reasonable state probability distribution [17,18]. In practice, the multi-state characteristics of a manufacturing system stem from the multi-state behavior of the manufacturing units it comprises. Therefore, a manufacturing system exhibiting such characteristics can be regarded as an MSMS composed of multi-state manufacturing units. These manufacturing units, in turn, can be modeled as traditional hierarchical MSSs.

In recent years, MSSs have been extensively studied for their ability to more accurately characterize the deterioration process of a system throughout its lifecycle compared to traditional binary-state models [19–21]. Due to their effectiveness in representing system degradation, MSS models have been widely applied in engineering practice, serving as a fundamental framework for evaluating the reliability [22], safety [23], and resilience [24] of complex systems. Han et al. [15] integrated the multi-state characteristics of manufacturing systems into their study; however, the primary emphasis was placed on remaining useful life estimation and maintenance decision-making. Importantly, the research did not present a dedicated reliability assessment methodology grounded in a robust multi-state modeling framework specifically tailored for manufacturing systems. Chen et al. [25] investigated the reliability assessment of manufacturing systems with multi-state characteristics. While they acknowledged the multi-state nature of manufacturing systems, their approach remained a conventional static reliability assessment method, lacking a dynamic perspective on system reliability evolution over time.

Traditional reliability assessment methods, which are based on large-sample statistical analysis theories, often face limitations in accurately capturing variations in failure patterns among individual components. The challenge of dynamically updating reliability has been a long-standing focus of research. In recent years, Scholars have examined reliability updates within the framework of structural reliability, using data such as load measurements [26] and test results [27] to improve the accuracy of reliability assessments. In modern production lines, a vast amount of real-time state monitoring data is readily available, either through expert observation or sensor-based monitoring. This data serves as a critical resource for dynamic reliability updates from a system reliability perspective [28,29], enabling more precise and adaptive reliability assessments. Liu et al. [30] applied a recursive Bayesian formula to update system reliability based on system-level observations, offering valuable insights into the application of Bayesian inference for reliability updates using observational data. Bayesian network (BN), developed based on Bayesian inference, offers a more intuitive and efficient framework for integrating observational information and performing probabilistic reasoning. Due to their powerful inferential capabilities, BNs have been extended to incorporate observational data from multiple levels [31] and multiple sources [32]. Additionally, they facilitate reliability parameter inference [33] and structure function learning [34], enhancing the dynamic assessment of MSS reliability. Luque and Straub [35] developed a modeling and computational framework based on DBNs to estimate and update the reliability of deteriorating structural systems. Their research established the groundwork for applying DBNs in reliability updates, particularly within structural reliability analysis. However, their approach did not integrate observational data for the dynamic updating of MSS. In practical applications, DBNs provide an effective framework for real-time reliability

updating in MSSs, positioning them as a promising tool for advancing dynamic reliability assessment methodologies.

Additionally, when updating system reliability, the epistemic uncertainty of observational information cannot be overlooked due to limitations in observation technology and the subjectivity of observer judgments [36,37]. Several epistemic uncertainty quantification methods are commonly used in reliability assessment, including fuzzy theory [38–40], probability box theory [41,42], and evidence theory [43–45]. Given that the system states of MSSs are discrete, evidence theory offers a natural advantage in characterizing imprecise state observations by representing them through state combinations. This characteristic makes it particularly suitable for representing the uncertainty associated with state observations in MSSs. Previous studies [46,47] have introduced methods for dynamically updating reliability using evidential observational data. However, prior research has predominantly focused on single-hierarchical systems, limiting its applicability to network-structured systems such as the production line-based manufacturing systems analyzed in this study. Modern manufacturing systems have become increasingly large and complex, with production lines emerging as a dominant form of industrial manufacturing. While previous methods have proven effective in updating the reliability of individual manufacturing units, they are not well-suited for the overall production line. To enable dynamic reliability updates for such systems, it is essential to develop a reliability assessment model that adopts a network-based perspective while effectively integrating information fusion techniques and dynamic reliability evaluation methodologies. Accordingly, novel approaches must be devised to address these challenges and enhance the accuracy and adaptability of reliability assessments.

3. Multistate flow network modeling for manufacturing systems

3.1. Multistate flow network

MFN is a mathematical framework used to model the flow of materials, information, and tasks as they traverse a network of interconnected nodes and edges [48]. In contrast to deterministic flow networks, which assume fixed values for flow and capacity, MFNs incorporate uncertainty in flow-related parameters, including stochastic variations in node or edge capacity, failure time, and processing duration. This framework enables the evaluation of the probability that flow will successfully traverse from source nodes to target nodes. Moreover, both the capacities and flow patterns within MFNs can dynamically change over time in response to observational data. This adaptability renders the MFN framework particularly suitable for time-evolving systems, such as those found in manufacturing and communication domains [49,50].

In the context of manufacturing systems, MFNs can be used to model the stochastic degradation behaviors of production lines, in which each individual manufacturing unit may fail randomly [51]. Since the degradation of manufacturing units is stochastic and the output of upstream manufacturing units affects the input of downstream manufacturing units, the work-in-process (WIP) quantities at each stage of a manufacturing system can be regarded as random variables once materials enter the system [10]. In this paper, the MFN-based reliability assessment model integrates rework and buffers, along with the multi-state characteristics of manufacturing units in manufacturing systems, to accommodate the real-world operating conditions of manufacturing systems.

This study investigates manufacturing systems under the following foundational assumptions. (1) To account for multi-state characteristics, the systems are modeled as MSMSs, in which multi-state manufacturing units are sequentially arranged in accordance with the processing order. (2) Rework is considered and is assumed to occur exclusively within designated manufacturing units. Each WIP item may undergo rework by the same unit no more than once, with the yield rate remaining constant throughout the rework process. (3) The capacities of distinct manufacturing units are regarded as statistically independent. (4) The state transitions of system components are assumed to follow a Markov process.

Suppose that an MSMS consists of n manufacturing units and m buffers, the MFN model of a MSMS is denoted by:

$$\mathrm{SFN} = \langle \boldsymbol{S}, \boldsymbol{B}, \boldsymbol{P}, \boldsymbol{T} \rangle. \tag{1}$$

where $\boldsymbol{S}$ is a set of states s_{ij}, s_{ij} is the jth state of the manufacturing unit m_i, c_{ij} represents the output capacity of manufacturing unit m_i when operating in state s_{ij}. $\boldsymbol{B} = \{b_1, b_2, \cdots, b_m\}$ is a set of buffers. $\boldsymbol{P} = \{p_1, p_2, \cdots, p_n\}$ is a set of yield rates of all the manufacturing units. $\boldsymbol{T} = \{t_1, t_2, \cdots, t_k\}$ is the set of production tasks, where $t_1, t_2, \cdots, t_k$ are different sub-tasks included in the MSMS.

Raw materials undergoing a specific process are graphically illustrated in Fig. 1 by the proposed SNF model. In the SNF framework, manufacturing units are represented as directed edges, whereas the WIP flowing through an edge signifies the completion of the corresponding process. Inspection stations are placed at both the upstream and downstream of manufacturing units to monitor the input and output quantities of WIP, and they are graphically represented as hollow circles as seen in Fig. 1 [8]. Consequently, in the MFN model of MSMSs, edges always direct from one hollow circle to another. Suppose that each inspection station is reliable, ensuring accurate inspection results, and the yield rate of each manufacturing unit corresponds to its predefined value.

As shown in Fig. 1, when I units of raw materials enter manufacturing unit i, the amount of WIP produced after processing by manufacturing unit m_i is O units. In the context of an MSMS, the system is considered reliable for a given task if the output quantity from the final manufacturing unit meets the task demand. In other words, the reliability of the MSMS under a specific task is defined as the probability that the final output quantity satisfies the required production demand. Given that manufacturing unit m_i operates with a yield rate of p, $O = p_i I$ units of WIP are generated after processing by manufacturing unit i.

In the case of rework, its graphical representation is delineated in Fig. 2. When rework is required at a specific manufacturing unit, the corresponding inspection station identifies whether the WIP output requires reprocessing, ensuring that it is efficiently routed either to the next manufacturing unit or for rework. As seen from Fig. 2 [8], rework occurs at manufacturing unit 2 (m_2), and the reprocessing starts from its preceding manufacturing unit (m_1), ensuring that defective products undergo correction before continuing to pass through the manufacturing system.

Since the yield rate of m_2 is p_2, its defective rate is given by $q_2 = 1 - p_2$. As illustrated in Fig. 2, after processing by manufacturing unit m_2, $p_1 p_2 I$ units of qualified products proceed to the next stage, while $p_1(1-p_2)I$ units of defective products undergo rework. Suppose that defective WIP can only be reworked once, the total output of a manufacturing unit with rework consists of two parts: (1) products qualified after the initial processing; (2) products qualified after a single rework. In general, rework in an MSMS is mathematically represented as [7]:

$$Rw = \left(rw_{f_1}^{m_1}, rw_{f_2}^{m_2}, \cdots, rw_{f_n}^{m_n} \right). \tag{2}$$

where $\left(rw_{f_1}^{m_1}, rw_{f_2}^{m_2}, \cdots, rw_{f_n}^{m_n} \right)$ represents the set of all rework processes.

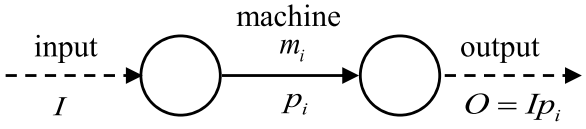


Fig. 1. Topological description of manufacturing unit inputs and outputs.

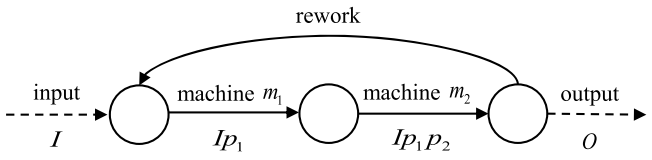


Fig. 2. Topological description of rework action in MSMS.

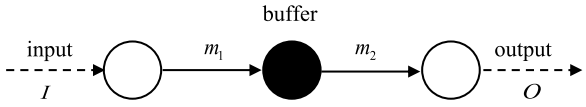


Fig. 3. Topological description of MSMS with a buffer.

In this equation, $rw_{f_i}^{m_i}$ indicates that manufacturing unit m_i involves rework, and the defective WIP is reprocessed beginning with the f_ith manufacturing unit preceding it.

In MSMSs, buffers are depicted as solid circles, as illustrated in Fig. 3 [8]. Buffers are placed between specific manufacturing units to regulate WIP flow. Infinite-capacity buffers cause no congestion and thus do not affect MSMS reliability. However, finite-capacity buffers may become congested when WIP exceeds their limits, disrupting production. Therefore, reliability assessment for MSMS with finite buffers must consider both whether unit states meet performance requirements and whether buffer capacity can sustain the required throughput.

3.2. Reliability assessment by multistate flow network model

In an MSMS, raw materials pass through multiple processing stages before becoming finished products. Reliability assessment begins with constructing an MFN model based on the system's actual structure by: (1) defining components (e.g., units, buffers, inspection stations) and their connections, and (2) identifying the multi-state parameters of the units. By the graphical representations of manufacturing units, buffers, and inspection stations introduced in Section 2.1, the topological structure of a complete MSMS is illustrated via Fig. 4 [8].

As shown in Fig. 4, the MSMS comprises five manufacturing units (m_1–m_5) with inspection stations at both input and output ends to monitor WIP in real time. Buffers are located after raw material entry, before product exit, and between m_2 and m_3 to regulate flow. Units are arranged sequentially by process, and varying yield rates cause WIP fluctuations, forming a WIP sequence. The minimal path (MP) is the minimum WIP needed to meet task demands across all processes. Under the framework of the MFN model, assessing the reliability of an MSMS lies in accurately identifying the MP [7].

Let $Pt = (m_1, m_2, \cdots, m_n, b_1, b_2, \cdots, b_l)$ denote a path for WIP from the raw material to the product, where $m_1, m_2, \cdots, m_n$ are manufacturing units in this MSMS and $b_1, b_2, \cdots, b_l$ are buffers. Let x_i denote the capacity of manufacturing unit m_i. The capacity vector of an MSMS with n manufacturing unit is denoted as $X = (x_1, x_2, \cdots, x_n)$. The *MPt* is defined as the minimal capacity vector that ensures the MSMS can produce enough products to meet the task demand, which can be expressed as [8]:

$$MPt = \mathbf{X}^{\min} = \left(x_1^{\min}, x_2^{\min}, \cdots, x_n^{\min}\right). \tag{3}$$

where $x_1^{\min}, x_2^{\min}, \cdots, x_n^{\min}$ is the minimal capacity of each manufacturing unit in this path. It is important to note that for different tasks, the MP varies, as it is task-dependent. Consequently, an MSMS may exhibit different reliability with respect to distinct tasks.This study defines the MSMS as a high-volume production system using standardized machinery and processes to produce identical or similar products. Since WIP defects may occur, the raw material input (I) must exceed the finished product output (O) to offset losses. Determining the required input quantity is essential for identifying the MP and evaluating system reliability under demand constraints.

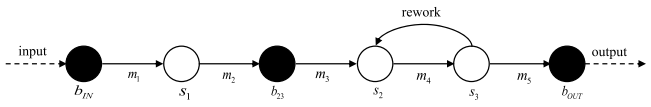


Fig. 4. Topological structure of an MSMS.

The quantity of input raw materials required to meet a specified output target can be determined through back-calculation. In an MSMS comprising n manufacturing units and operating without rework, the relationship between the input and output quantities is expressed as follows [7]:

$$O = Ip_1p_2\cdots p_n, \tag{4}$$

where $p_1, p_2, \cdots, p_n$ are the yield rate of each manufacturing unit. I is the input quantity of raw materials. When the task demand d is known, the amount of input raw materials can be computed by [7]:

$$I = \frac{d}{p_1p_2\cdots p_n}. \tag{5}$$

When rework is considered, the output of the MSMS consists of two components: (1) Products qualified during the initial production process and (2) Products that pass quality standards after undergoing rework. Under these two scenarios, the relation between input and output is given by:

$$O = Ip_1p_2\cdots p_n + Rw_{O1} + Rw_{O2} + \cdots + Rw_{On} \tag{6}$$

where $Rw_{O1}, Rw_{O2}, \cdots, Rw_{On}$ are the qualified products output by rework $rw_{f_1}^{m_1}, rw_{f_2}^{m_2}, \cdots, rw_{f_n}^{m_n}$ respectively. For rework $rw_{f_i}^{m_i}$, the qualified products output by this rework is:

$$Rw_{Oi} = Ip_1p_2\cdots p_{m_{ri}-1}\left(1-p_{m_{ri}}\right)p_{m_{ri}-f_i}\cdots p_n. \tag{7}$$

Furthermore, the input raw materials can be derived by:

$$I = \frac{O}{p_1p_2\cdots p_n + \sum_{i=1}^{n} p_1p_2\cdots p_{m_{ri}-1}\left(1-p_{m_{ri}}\right)p_{m_{ri}-f_i}\cdots p_n} \tag{8}$$

Based on Eqs. (6), (7), and (8), if the demand is determined, which means $O = d$, the amount of input raw materials can be obtained when rework is considered. Taking the MSMS in Fig. 4 as an example for illustration. Since the rework of manufacturing unit m_4 is considered in this MSMS, the defective rate of manufacturing unit m_4 is $(1 - p_4)$. This means that $Ip_1p_2p_3(1-p_4)$ units of WIP will re-enter manufacturing unit m_3 for rework after being processed by manufacturing unit m_4. Therefore, the relation between output and input is as following:

$$O = Ip_1p_2p_3p_4p_5 + Ip_1p_2p_3(1-p_4)p_4p_5. \tag{9}$$

According to the task demand, if the task demand is to produce d units of products, the relation between the amount of raw material input and task demand can be derived as:

$$I = \frac{d}{p_1p_2p_3p_4p_5 + Ip_1p_2p_3(1-p_4)p_4p_5}. \tag{10}$$

The input flow for each manufacturing unit is determined after calculating the required amount of input raw materials. The MP, which represents the minimal path to meet the task demand, can also be derived according to the minimal amount of input raw materials. Due to the rework in the manufacturing process, the minimal capacity of the manufacturing unit can be divided into two parts: the initial part and the rework part. According to the rework $Rw = \left(rw_{f_1}^{m_1}, rw_{f_2}^{m_2}, \cdots, rw_{f_n}^{m_n}\right)$

defined before, the minimal capacity $x_j^{\min}$ of manufacturing unit m_j can be calculated as following:

$$X_j^{\min}=\begin{cases} I\prod_{x=1}^{j}p_x & ifj<m_i-f_i \\ I\prod_{x=1}^{j}p_x+\sum_{i=1}^{n}\left[I\prod_{x=1}^{j}p_x\left(1-p_{m_i}\right)\prod_{x=m_i-f_i}^{j}p_x\right] & ifj\in[m_i-f_i,m_i] \\ I\prod_{x=1}^{j}p_x+\sum_{i=1}^{n}\left[I\prod_{x=1}^{j}p_x\left(1-p_{m_i}\right)\prod_{x=m_i-f_i}^{m_i}p_x\right] & ifj>m_i \end{cases}. \tag{11}$$

In Eq. (11), I is the amount of input raw materials obtained according to Eq. (8), and p is the yield rate. If $j<m_i-f$, the minimal capacity of m_j is determined solely by the initial part and is not affected by rework. If $j>m_i-f$, minimal capacity of m_j is the sum of the initial part and the rework part. The MSMS system in Fig. 4 is still used as an example for explanation. Rework rw_1^4 is the only rework action in this MSMS, such that the minimal capacity $x_4^{\min}$ and $x_5^{\min}$ should be conculcated by adding the reworked WIP. According to Eq. (11), the minimal capacity of each manufacturing unit in the MSMS in Fig. 4 can be calculated as follows:

$$\begin{aligned} x_1^{\min}&=Ip_1\\ x_2^{\min}&=Ip_1p_2\\ x_3^{\min}&=Ip_1p_2p_3\\ x_4^{\min}&=Ip_1p_2p_3p_4+Ip_1p_2p_3p_4(1-p_4)\\ x_5^{\min}&=Ip_1p_2p_3p_4p_5+Ip_1p_2p_3p_4p_5(1-p_4) \end{aligned}$$

Suppose $Pt^*=\left(Pt_1^*,Pt_2^*,\cdots,Pt_k^*\right)$ is the set of feasible paths that can meet the production requirements and buffer constraints. Since the capacities of different manufacturing units are assumed to be statistically independent, the probability of a path Pt_i^* is the product of the state probabilities of each manufacturing unit in the path. If the probability of path Pt_i^* is r_i^*, the reliability of the MSMS can be calculated by:

$$R=1-\left(1-r_1^*\right)\left(1-r_2^*\right)\cdots\left(1-r_i^*\right). \tag{12}$$

In summary, the reliability calculation of the manufacturing system in this study can be divided into the following key steps:

Step 1: Construct the MFN-based reliability assessment model for the MSMS by determining the number and arrangement of manufacturing units, the placement and capacity of buffers, and the reworks.

Step 2: Calculate the input quantity of raw materials according to task demand based on Eqs. (6)-(8).

Step 3: Compute the minimal input flow based on the amount of input materials obtained before, and then determine the MP according to Eq. (11).

Step 4: Define the states of manufacturing units in the MSMS according to the performance degradation and varying production efficiency. Then, establish the state probability distribution for each manufacturing unit by using an appropriate probability distribution mode.

Step 5: Identify the set of all feasible paths $Pt^*=\left(Pt_1^*,Pt_2^*,\cdots,Pt_k^*\right)$.

Step 6: Given the MP and the state probability distributions of the manufacturing units, calculate the probability of each feasible path. Finally, calculate the reliability of the MSMS using Eq. (12) by aggregating the probabilities of all feasible paths.

In this method, MSMS reliability evaluation is based on the state probability distribution of each unit, typically obtained from a probabilistic model using historical failure data. This conventional static approach mainly assesses specific product categories. To improve precision and adaptability, the following sections present a dynamic reliability assessment method incorporating observational data.

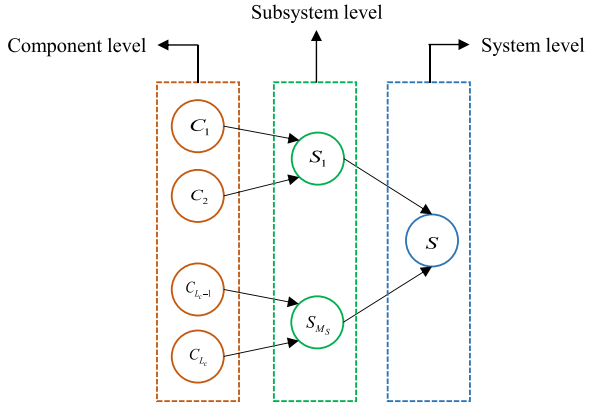


Fig. 5. BN of the hierarchical MSS.

4. Dynamic reliability assessment for manufacturing units

4.1. Dynamic Bayesian network of manufacturing units

To capture gradual degradation and intermediate states in real-world manufacturing, this study adopts the MSMS framework, where multi-state behavior stems from individual units. Unlike binary models, the MSS approach accommodates multiple performance levels, enabling more realistic reliability analysis, especially for units with partial functionality or degradation rather than complete failure.

In a manufacturing system, each unit—such as a high-end CNC machine—can be modeled as a hierarchical MSS with multi-state components, where both the system and its components have a finite number of discrete performance states. Since this article focuses on updating reliability based on observation information, the BN method is introduced. This method effectively facilitates probabilistic inference among different nodes, enabling the update of probabilities of other nodes based on information from a given node.

A BN typically comprises a topology and a conditional probability table (CPT). The topology is represented as a directed acyclic graph composed of a set of nodes $\Omega=\{X_1,X_2,\cdots,X_n\}$ and directed edges connecting them. If there exists a directed arc from X_j to X_i, X_j is called the parent node of X_i, denoted as $\mathrm{Pa}(X_i)$, X_i is a child node of the parent node X_j. A child node X_i can have multiple parent nodes, and a parent node X_j may connect to multiple child nodes as well. The topology of a BN reflects the causal relationships between nodes, while the CPT quantitatively defines the strength of these relationships. Using Bayes' theorem, the joint probability distribution of the network can be expressed as [40]:

$$\Pr\{\Omega\}=\prod_{i=1}^{n}\Pr\{X_i|\mathrm{Pa}(X_i)\}, \tag{13}$$

where $\Pr\{\Omega\}$ is the joint probability distribution of Ω, and $\mathrm{Pa}(X_i)$ is the parent nodes of node X_i. For each node, the marginal probability distribution can be derived by [40]:

$$\Pr\{X_i\}=\sum_{\Omega/X_i}\Pr\{\Omega\}. \tag{14}$$

BNs are able to represent a system as an interconnected network, effectively capturing dependencies between components. They have been extensively utilized in modeling systems as they seamlessly integrate various factors, including interactions, uncertainties, and outcomes, while also facilitating the analysis of trade-offs [52]. Under the assumptions of binary-state system reliability theory, the system reliability can be calculated using the following formula [40]:

$$R(S)=\Pr\{S=1\}. \tag{15}$$

In this study, each manufacturing unit is regarded as a hierarchical MSS with multi-state components. Within the framework of BNs, components, subsystems, and the entire system can be represented as nodes.

Directed edges are introduced between nodes to capture dependency relations among different physical levels of the system. Take the hierarchical MSS illustrated in Fig. 5 as an example; this system consists of L_C components, which are further structured into M_S subsystems. As shown in Fig. 5, the BN of the specific MSS can be represented by $L_C + M_S + 1$ nodes with directed edges. The nodes in the BN can be expressed as $\Omega = \{X_1, X_2, \cdots, X_n\} = \{C_1, C_2, \cdots, C_{L_C}, S_1, S_2, \cdots, S_{M_S}, S\}$, where $C_l (l = 1, 2, \cdots, L_C)$ represents component l, $S_m (m = 1, 2, \cdots, M_S)$ represents subsystem m and S is a node for the entire system.

The joint probability distribution and marginal probability distribution can be obtained using Eq. (13) and Eq. (14), respectively. As manufacturing units represented by nodes have more than two states, if c_S is the performance level of the system, the reliability of the entire system can be determined by defining the system demand ω, expressed mathematically as:

$$R(S) = \Pr\{c_S > \omega\}. \tag{16}$$

Since this study considers the time-varying reliability of each hierarchical level within a manufacturing unit, the BN can be extended along the time horizon to capture the dynamic evolution of reliability over time. By incorporating the time horizon, the BN in Fig. 5 extends to a DBN, where sequentially linked time slices capture system state evolution. Each slice models the MSMS at a given time, while temporal links represent unit degradation. This structure integrates static dependencies within slices and dynamic transitions between them, enabling unified analysis of long-term system reliability. As for a hierarchical MSS, the DBN model is composed of a set of nodes, denoted as $\Omega = \{C_l(t), S_m(t), S(t)\}$, where $(l = 1,2,\cdots,L_C; m = 1,2,\cdots,M_S; t = 1,2,\cdots,T)$. The $C_l(t)$, $S_m(t)$, and $S(t)$ represent component l, subsystem m and the entire system at the time slice t, respectively. T is the time duration of interest. Fig. 6 illustrates the graphical structure of a hierarchical MSS.

Likewise, the joint probability distribution for all nodes in the DBN can be represented as the product of the conditional probabilities of all nodes, conditioned on their respective parent nodes [40]:

$$\Pr\{\Omega\} = \prod_{t=0}^{T} \prod_{i=1}^{n} \Pr\{X_i(t) | \mathrm{Pa}(X_i(t))\}, \tag{17}$$

where $\mathrm{Pa}(X_i(t))$ is the set of parent nodes of $X_i(t)$.

This study assumes component degradation follows a Markov process, where transitions depend only on the current state. This widely accepted assumption reflects the memoryless nature of many physical processes and enables efficient state distribution computation via one-step transition matrices (TMs), supporting the DBN's temporal modeling. By the established DBN model, the state probability combinations of components corresponding to the states of subsystems and the entire system can be determined, ultimately leading to the state probability distribution of the entire system. Once this distribution is obtained, system reliability under different demand scenarios can be calculated. Building on this foundation, in the DBN model of a hierarchical MSS, the transitions between different time slices effectively represent the degradation of components over time, allowing for a dynamic evaluation of system reliability as it evolves.

4.2. Multi-level evidential observations

For a specific running manufacturing unit in a manufacturing system, the stochastic nature of degradation presents a challenge for traditional reliability assessment methods, which only concern the reliability of a population of identical systems, and the system-to-system reliability variation in the operational stage cannot be accurately reflected. Since DBNs capture the dynamic evolution of nodes over time, they provide a general framework for dynamically updating the reliability of a specific individual system by using observational data

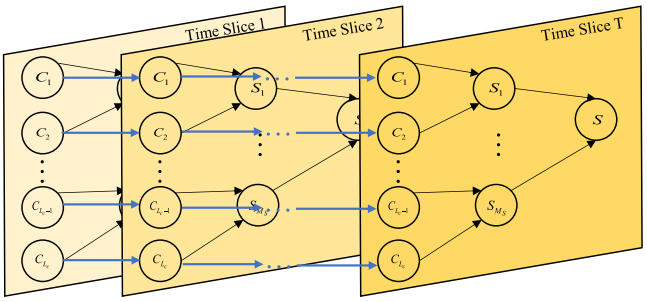


Fig. 6. DBN of the hierarchical MSS.

collected at different time points. Nevertheless, the observational data collected by integrating real-time monitoring and expert assessments inevitably contain epistemic uncertainty in engineering practice.

Evidence theory, proposed by Dempster and Shafer [53,54], is a structured and adaptable approach for representing epistemic uncertainty in MSMSs. Unlike fuzzy logic, which requires continuous membership functions, or interval probability, which only provides probability bounds, evidence theory assigns belief masses to both singleton and composite state subsets, effectively handling discrete states and ambiguous information. Compared with imprecise Markov models, it avoids complex interval transition matrices and integrates naturally with hierarchical Bayesian frameworks. Its ability to fuse expert judgment and sensor observations via Dempster's rule makes it particularly suitable for the dynamic reliability assessment in this study.

In the evidence theory, the framework of discernment (FoD) Θ is a finite and complete set of hypotheses for a variable X [40]. If the FoD Θ is represented as $\Theta = \{x_1, x_2, \cdots, x_n\}$, all the possible subset propositions under the FoD Θ form a power set as follows [36]:

$$2^{\Theta} = \{\varnothing, x_1, x_2, \cdots, x_n, [x_1, x_2], [x_1, x_3], \cdots, \Theta\}, \tag{18}$$

where $\varnothing$ is the empty set and each element S of this set is corresponding to the possible propositions. Specifically, for the state observation focused on in this study, element $x_1, x_2, \cdots, x_n$ are the states of the observed components, subsystems or system. The basic belief assignment, commonly referred to as the mass function $m(S)$, assigns values from the interval $[0, 1]$ to the power set of the FoD Θ. The mass function $m(S)$ can be interpreted as a measure of the degree of support for the proposition S. The mass function satisfies the following normalization axioms [43]:

$$m(\varnothing) = 0, \tag{19}$$

$$\sum_{S \subseteq \Theta} m(S) = 1. \tag{20}$$

The belief function $Bel(S)$ and the plausibility function $Pl(S)$, defined as follows, represent the lower and upper bounds of confidence for a given proposition, respectively, quantifying its minimum commitment and maximum plausibility [43]:

$$Bel(S) = \sum_{T \subseteq S} m(T), \tag{21}$$

$$Pl(S) = \sum_{S \cap T \neq \varnothing} m(T). \tag{22}$$

While $[Bel(S), Pl(S)]$ represents the degree of epistemic uncertainty associated with S. Using the evidential observation information applied in this study as an example, if the hypothesis S is a single-element hypothesis, the size of the interval $[Bel(S), Pl(S)]$ represents the degree of imprecision in observing the system belonging to a specific state. This study focuses on the dynamic reliability updating, where the foundation

for achieving dynamic assessment lies in the merging of different state information to complete the updating process. The rich set of fusion rules in evidence theory provides strong support for the implementation of the proposed method [55]. As one of the most popular fusion rules in the evidence theory, the Dempster rule of combination (DRC) "$\oplus$" can be used to merge different information, and is defined as [36]:

$$(m_1 \oplus m_2)[C] = \begin{cases} \dfrac{\sum_{A\cap B=C} m_1(A)m_2(B)}{1-k} \text{if} C \subseteq \Theta \\ 0 \text{if} C = \varnothing \end{cases}, \tag{23}$$

where $k = (m_1 \cap m_2)[\varnothing] = \sum_{A\cap B=\varnothing} m_1(A)m_2(B)$ is called the degree of conflict. This rule applies to the merging of evidence bodies as well as the fusion of evidence bodies with probability values, as probability values can essentially be regarded as a specific type of evidence value.

In this study, manufacturing units are modeled as MSSs with multi-state components, and observation data from multiple physical levels can be used to update the system reliability. Observations at each physical level are represented using mass functions to account for epistemic uncertainty. These mass functions provide a tool for quantifying epistemic uncertainty, effectively capturing and representing uncertain observations by incorporating subsets of state information. At time instant t, the system-level, subsystem-level, and component-level evidential observations can be, respectively, denoted as $\widetilde{\mathbf{o}}^{S}(t)$, $\widetilde{\mathbf{o}}^{S_m}(t)(m \in 1,2,\cdots,M_S)$, and $\widetilde{\mathbf{o}}^{C_l}(t)(l \in 1,2,\cdots,L_C)$.

For the evidential observation of hierarchical MSSs with multi-state components, the FoD of each node must encompass all possible states of that particular node. Theoretically, there are 2^{Θ^S} possible subset propositions under FoD Θ^S. Similarly, $2^{\Theta^{S_m}}$ and $2^{\Theta^{C_l}}$ possible subset propositions are under Θ^{S_m} and Θ^{C_l}. These subsets represent all the possible combinations of states that can occur for each node within the system, providing a basis for uncertainty quantification in hierarchical MSSs with multi-state components. More specifically, the evidential observations of components at any time t are denoted as $\widetilde{\mathbf{o}}^{C_l}(t) = \mathbf{m}^{C_l}(t)$. The corresponding FoD of $\mathbf{m}^{C_l}(t)$ is $\Theta^{C_l} = \{0,1,2,\cdots,N_{C_l}\}$, where $\{0,1,2,\cdots,N_{C_l}\}$ is the set of the lth component's state. Similarly, $\widetilde{\mathbf{o}}^{S_m}(t)$ and $\widetilde{\mathbf{o}}^{S}(t)$, the evidential observation of subsystems and the system can also be defined accordingly

In practical engineering, eliciting the mass function for such observations is fundamentally equivalent to eliciting the contour function. As the contour function is applied exclusively to singleton sets rather than power sets, it offers a more intuitive approach for probabilistic reasoning.

4.3. Dynamic reliability assessment by evidential observations

This dynamic reliability assessment of manufacturing units can be addressed through a two-step framework: (1) reliability assessment based on historical failure data, and (2) dynamically updating the reliability by merging state observational data. The prior reliability of manufacturing units can be calculated by building the DBN model and determining the one-step state TMs of components within the manufacturing units. Additionally, the reliability can be dynamically updated based on the following way.

The integration of evidential observations is carried out using the DRC framework, which provides an effective mechanism for combining information derived from multiple bodies of evidence. Since probability values can be interpreted as a specific case of evidence bodies, the DRC framework is also applicable to the fusion of probability values with evidential information. Within the context of the MSS dynamic reliability updating addressed in this study, the system state probability distributions obtained from statistical data are treated as probability values, whereas imprecise observational data are modeled as evidence bodies. The application of DRC enables the effective merging of these two distinct forms of information.

Taking the merging and updating process based on component-level observations as an example, the evidential observation for the lth component node is associated with a FoD represented by $\Theta^{C_l} = \{0,1,2,\cdots,N_{C_l}\}$. All possible subset propositions within this FoD are represented as follows:

$$2^{\Theta^{C_l}} = \{\varnothing, 0, 1, 2, \cdots, N_{C_l}, [0,1], [0,2], \cdots, [0,1,2,\cdots,N_{C_l}]\}, \tag{24}$$

where $\varnothing$ is the empty set. The mass function $m^{C_l}(t)$ of the power set can be determined according to the imprecise observation. The goal is to integrate the probability distribution of the component node derived from historical failure data, denoted as $\mathbf{p}^{C_l}(t)$, with the evidential observations, denoted as $\mathbf{m}^{C_l}(t)$, at time instant t. The previously discussed DRC is employed to facilitate this information merging. For a component node, the state probability distribution obtained from historical statistics and the evidential observations can be combined as follows:

$$\mathbf{m}^{C_l}_{updated}(t) = \left[\mathbf{p}^{C_l}(t) \oplus \mathbf{m}^{C_l}(t)\right][A] = \begin{cases} \dfrac{\sum_{B\cap C=A} p_B^{C_l}(t) m_C^{C_l}(t)}{1-k} \text{if} A \subseteq \Theta \\ 0 \text{if} A = \varnothing \end{cases}, \tag{25}$$

where $\mathbf{m}^{C_l}_{updated}(t)$ is the mass function of the lth component at time instant t after refreshing. Since $\mathbf{p}^{C_l}(t)$ is a probability measure rather than an evidential body, B is limited to singletons. k is the degree of conflict, and $k = \left(\mathbf{p}^{C_l}(t) \cap \mathbf{m}^{C_l}(t)\right)[\varnothing] = \sum_{B\cap C=\varnothing} p_B^{C_l}(t) m_C^{C_l}(t)$.

In engineering practice, constructing a complete and reasonable mass function can be challenging, as it requires assigning beliefs to all subset propositions. Instead, the contour function $\mathbf{pl}^{X_i}(t) = \left(\mathrm{pl}_0^{X_i}(t), \mathrm{pl}_1^{X_i}(t), \ldots, \mathrm{pl}_{\Theta^{X_i}}^{X_i}(t)\right)$ under the FoD is a more practical and preferred approach for representing the uncertainty of evidential observations. Engineers often adopt this format to convey their evidential knowledge since experts are typically asked to express beliefs only on singletons, making the process more intuitive and manageable. In this study, the mass function is replaced by the following contour function [40]:

$$\mathrm{pl}_A^{X_i}(t) = \begin{cases} 1 & \text{If expert believes node} X_i \text{ is in state } A, \\ \mu & \text{For other states of node } X_i \end{cases} \quad A \in \Theta^{X_i}, \tag{26}$$

where $\mu \in [0,1]$ is an indicator used to quantify the degree of imprecision in the observation. Specifically, when $\mu = 0$, it indicates that the observation is precise; when $\mu = 1$, it signifies that no valuable observation information has been obtained; and when $\mu \in (0,1)$, it suggests that the system is considered to be in a state with a certain degree of likelihood. With the increment of the parameter μ, the epistemic uncertainty gradually increases [39]. In practical settings, engineers are not required to provide exact probabilities for component states. Instead, qualitative evaluations are performed based on diagnostic indicators such as vibration, temperature, or control feedback, using linguistic descriptors such as "likely," "possible," or "unlikely." These descriptors are subsequently mapped to plausibility values to define the contour function. For standardization purposes, the following mapping is adopted: "very likely" = 0, "likely" = 0.2, "possible" = 0.4, "unlikely" = 0.7, and "very unlikely" = 0.9. For instance, if a component is assessed as "possible" in state 0, the corresponding contour function assigns a value of 1 to state 0 and 0.4 to all other states.

While the proposed method provides a structured framework for the dynamic reliability assessment of manufacturing units, several limitations must be acknowledged. First, constructing and updating conditional probability tables in DBNs becomes computationally intensive in large-scale systems involving numerous multi-state components. Second, the accuracy of the updated reliability estimates is highly dependent on the quality and availability of observational data; sparse or conflicting observations may result in unstable outcomes. These

limitations also indicate potential directions for future research.

5. Dynamic reliability assessment for the multi-state manufacturing system

5.1. From manufacturing unit reliability to manufacturing system reliability

As discussed in Section 3, the manufacturing system in question exhibits multi-state characteristics, where each manufacturing unit can be modeled as an MSS. Since the manufacturing system under study is a single-sequence flow system composed of multiple interdependent manufacturing units, the manufacturing system can be collectively regarded as an MSMS.

Section 4 introduced a dynamic reliability updating method for hierarchical MSSs using imprecise observational data. This approach updates unit state probabilities in real time and assesses overall MSMS reliability by aggregating probabilities along all feasible MFN paths. The DBN models temporal state evolution, while the MFN captures system structure and computes task reliability based on updated states, linking local uncertainty to overall performance dynamically.

Fig. 7 illustrates the proposed framework, where the dynamic updating of the MSMS reliability fundamentally depends on accurately capturing the reliability evolution of each manufacturing unit. By integrating the MFN-based network-level modeling and the MSS-based unit-level dynamic assessment, this study extends dynamic reliability methodologies from hierarchical structures to network-structured manufacturing systems. Here, the MSMS framework is selected for this study because it offers the flexibility to capture gradual machine degradation and discrete performance levels, which are often observed in real-world production environments. Unlike traditional binary models, the MSMS representation can accommodate intermediate manufacturing unit states, providing a more realistic understanding of system performance. The DBN model is used for their ability to propagate probability updates across the network, enabling dynamic and comprehensive reliability inference. The MFN model captures the structural and functional relationships between manufacturing units from a network perspective. Imprecise state observations are chosen because they are commonly available and easily obtained in engineering practice, making the method practical and applicable.

5.2. The proposed dynamic reliability assessment method

The central principle of the proposed dynamic reliability assessment methodology involves updating the state probability distributions of manufacturing units within the MSMS by incorporating observational data, particularly imprecise observations. These updated state distributions are subsequently utilized to calculate the reliability at the network level.

Due to the presence of epistemic uncertainty, the observation data acquired from experts or sensors is imprecise. The methodology for merging imprecise observation data has been introduced in Section 4,

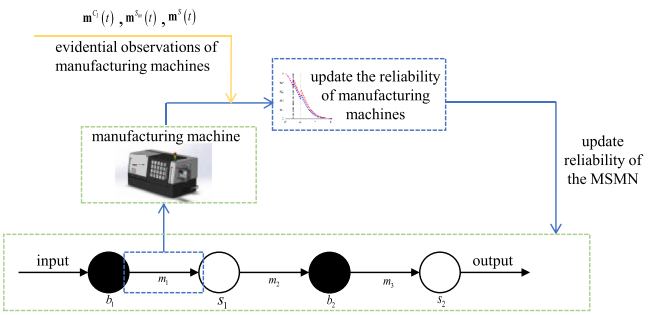


Fig. 7. An illustration of manufacturing system reliability updating.

providing a framework for effectively integrating uncertain information into the dynamic reliability assessment process. The updated state probability distributions of manufacturing unit m_i can be denoted as a vector of $\mathrm{P}^{\wedge m_i} = \left(p^{\wedge m_i}_1, p^{\wedge m_i}_2, \cdots, p^{\wedge m_i}_{N_{m_i}}\right)$, where $p^{\wedge m_i}_j = \Pr\{\text{state of} m_i \text{is} S_{ij}\}$ and $j \in (1, 2, \cdots, N_{m_i})$. Compared with the original probability distributions $\mathrm{P}^{m_i} = \left(p^{m_i}_1, p^{m_i}_2, \cdots, p^{m_i}_{N_{m_i}}\right)$ which were estimated purely from historical failure data, the updated probability distributions will theoretically be closer to the true state probability distribution of a specific observed manufacturing unit as the observation data has been incorporated.

Once the unit-level state probabilities are updated, the network-level reliability can be recalculated by evaluating the total probability of all feasible operational paths P^*_i. According to the updated state probability distributions, r^*_i, the updated probability of feasible path P^*_i, can be updated to $r^{\wedge *}_i$ as well. Hence, the reliability of the MSMS can be obtained by:

$$R = 1 - \left(1 - r^*_1\right)\left(1 - r^*_2\right)\cdots\left(1 - r^*_i\right). \tag{27}$$

Building on the introduction to the MSMS dynamic reliability updating method, the dynamic reliability updating process is implemented through the following steps:

Step 1: Establish the MFN-based reliability assessment model of the MSMS. Define manufacturing unit processing sequences, identify all manufacturing units and buffers, and describe the logical structure of the system.

Step 2: Specify the performance levels (states) for each manufacturing unit and estimate the initial state probability distributions based on historical failure or degradation data.

Step 3: For a given production task, determine the required output d and calculate the task reliability of the MSMS under this demand d base on the reliability assessment model established in Step 1 and the manufacturing unit state probability distributions obtained in Step 2.

Step 4: Collect the evidential state observation information for each manufacturing unit in the MSMS at selected observation time points. For each manufacturing unit, the most probable state and its corresponding contour function are provided by sensors or expert judgment.

Step 5: Fuse the real-time state observation date with the prior state probabilities, following the rules introduced in Section 3.3, ensuring an updated and refined state probability distribution for each manufacturing unit.

Step 6: Recalculate the reliability of the MSMS by evaluating the updated probabilities of all feasible paths under the current task demand. Repeat the procedure dynamically over time and for different tasks as needed.

This study provides a flexible and adaptive framework for dynamic reliability assessment that can accommodate multi-level, imprecise, and time-varying information. After each dynamic reliability assessment using observational data, the evaluation results are updated accordingly. The outcomes obtained by this method are real-time and continuously revised. Compared with traditional methods, traditional DBN-based reliability models typically rely solely on historical transition probabilities and lack mechanisms for real-time updating based on new observations. In practical engineering scenarios, accurately observing all hierarchical states of each unit within a manufacturing system remains challenging. However, acquiring imprecise state information for selected units and specific levels is comparatively more achievable. Therefore, the method proposed in this study, which updates overall system reliability based on locally imprecise observations, demonstrates wide applicability in contexts where precise state data for all manufacturing units is not attainable. However, in the DBN used for system reliability inference, the computational cost of exact inference grows approximately as $O\left(T{\cdot}2^k\right)$, where T is the number of time steps and k is the maximum number of parent nodes for any single node. This observation underscores that although the DBN exhibits linear expan-

sion over time, the size of the CPTs increases exponentially with the number of parent variables. This exponential growth considerably elevates the computational complexity of probabilistic inference as system dependencies become more intricate. Addressing this computational challenge represents a critical direction for future research.

6. Illustrative example

6.1. Aluminum alloy wheels production line

In recent years, the rapid advancement of new energy vehicles has stimulated significant growth in the automotive industry, further accelerating its global expansion. As a fundamental component in vehicle manufacturing, aluminum alloy wheels are installed on the inner walls of tires, providing structural support and ensuring secure attachment [56]. This paper employs the aluminum alloy wheel production line as an illustrative example to demonstrate the effectiveness of the proposed MSMS dynamic reliability assessment method.

Within the framework of Industry 4.0, automated production lines for automotive wheels have been progressively developed and implemented, enhancing manufacturing efficiency and precision. The illustrative example presented in this section is derived from an actual aluminum alloy wheel manufacturing line. The reliability-critical subsystems considered in this study, including CNC machining centers, spindles and feed mechanisms, control units, and auxiliary devices, are representative of components commonly employed in automated wheel production. The manufacturing process of aluminum alloy wheels, transitioning from raw blanks to finished products, comprises three sequential machining stages, as shown in Fig. 8. This production line adopts a cost-effective configuration of manufacturing equipment, consisting of two lathes and one machining center. The lathes include a horizontal lathe and a vertical lathe equipped with double turrets, while the machining center is a vertical machining unit.

During production, the processing efficiency of manufacturing units gradually degrades, leading to multi-state processing levels. In this study, the aluminum alloy wheel production line is modeled as an MSMS, where each manufacturing unit is treated as a hierarchical MSS. Given the complexity of the machining processes involved in aluminum alloy wheel production, rework is specifically considered for the machining center. Reworked products are directly reprocessed from this stage without requiring the repetition of preceding procedures. Additionally, two buffers are incorporated to maintain a stable production flow. Following the MFN-based modeling approach outlined in Section 2, the resulting model of the automated production line for automotive wheels is presented in Fig. 9.

To accurately assess the reliability of this production line, it is essential to establish an MFN-based reliability assessment model. As a complex system with degradation and a long lifecycle, the three manufacturing units in the production line can be approximated as MSSs with multi-state components. Their states are classified based on the degree of degradation in production performance. Since the performance of the production line degrades over time, the state probability distribution of each manufacturing system evolves accordingly. Therefore, it's worth studying the task-based reliability of the MSMS at specific time instants under certain production demand conditions. The reliability updating method proposed in this study enables accurate assessment of production line reliability and provides recommendations for further improvements.

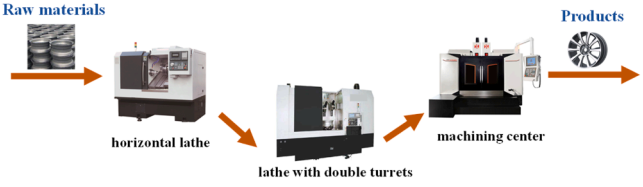


Fig. 8. The manufacturing process of aluminum alloy wheels.

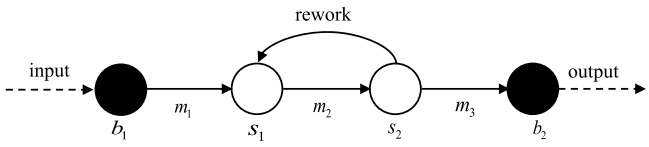


Fig. 9. The MFN model of aluminum alloy wheel production line.

6.2. Dynamic Bayesian network of alloy wheels production line

To update the reliability of the aluminum alloy wheel production line, it is necessary to achieve dynamic reliability updating of the manufacturing systems within it. Consequently, the DBN model of the CNC machines in the production line must first be established. In this illustrative example, although the manufacturing units in the aluminum alloy production line serve different functions, they all belong to CNC machines and share similar structural compositions. Based on the structure and components of CNC machines, the BN topology, as shown in Fig. 10, is constructed. This topological structure applies to all manufacturing units in the production line. The differences among various types of CNC machines are reflected in the state definitions of systems and components, as well as the one-step TM of components.

As depicted in Fig. 10, the CNC machine system is structured into four subsystems comprising a total of ten components. Subsystem 1, the machine body, includes the bed, spindle box, feed mechanism, and tool-changing device. Subsystem 2, the numerical control system, consists of the CNC, PLC, and display unit. Subsystem 3, the servo and measurement system, incorporates the spindle drive unit, feed drive unit, and measurement elements. The state definitions corresponding to various CNC machines are detailed in Table 1, while the conditional probability table is provided in Appendix 1. The node states outlined in Table 1 reflect distinct capacity levels, which serve as the foundation for subsequent computations of task-based production line reliability.

To support dynamic reliability modeling, each component in the CNC machine tool system is classified into discrete states based on observable physical degradation and failure criteria. Components such as the bed, spindle box, tool-changing device, display unit, and measurement element are modeled using a two-state framework. State 0 (Normal) signifies that the component is operating within acceptable parameters. For instance, the bed retains its structural integrity and geometric precision, the spindle box exhibits no abnormal vibration or thermal behavior, the display and sensors deliver accurate feedback, and tool-changing operations are completed without failure. State 1 (Failure) is assigned when any of these components become non-functional or unsafe for continued operation, such as major structural

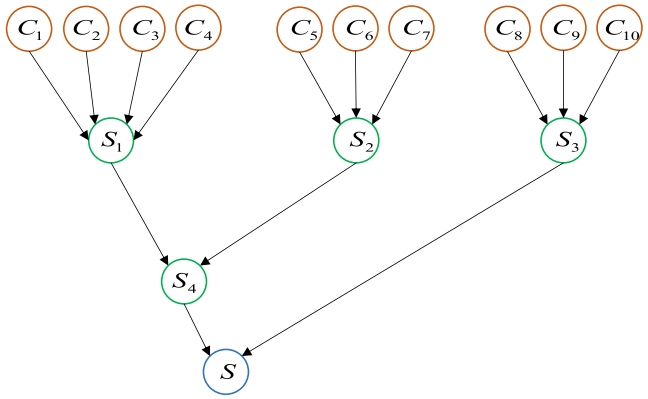


Fig. 10. The topology of CNC machines' BN model.

Table 1
The states of each CNC machine.

Node	State of horizontal lathe (m_1)	State of vertical lathe (m_2)	State of machining center (m_3)
S	0,1,2,3	0,1,2,3	0,1,2,3,4
S_1	0,1,2	0,1,2	0,1,2
S_2	0,1,2	0,1,2	0,1,2
S_3	0,1,2	0,1,2	0,1,2
C_1 (bed)	0,1	0,1	0,1
C_2 (spindle box)	0,1	0,1	0,1
C_3 (feed mechanism)	0,1,2	0,1,2	0,1
C_4 (tool-changing device)	0,1	0,1	0,1
C_5 (CNC)	0,1,2	0,1,2	0,1,2
C_6 (PLC)	0,1,2	0,1,2	0,1,2
C_7 (display unit)	0,1	0,1	0,1
C_8 (spindle drive unit)	0,1,2	0,1,2	0,1,2
C_9 (feed drive unit)	0,1,2	0,1,2	0,1,2
C_{10} (measurement element)	0,1	0,1	0,1
	Note: State 0 represents the best condition.		

deformation of the bed, spindle jamming, failure of the tool-changing actuator, unresponsive display, or sensor signal loss. Other components, including the feed mechanism, CNC controller, PLC, spindle drive unit, and feed drive unit, are modeled using a three-state representation to capture more detailed degradation behavior. State 0 (Normal) indicates standard operational performance, such as accurate feed motion, prompt controller responses, and stable drive functions. State 1 (Degraded) characterizes reduced functionality, including minor positioning inaccuracies due to backlash in the feed mechanism, occasional delays in the PLC or CNC response, or non-critical irregularities in drive power output. State 2 (Failure) represents complete functional loss, such as servo control failure, system crashes in the logic controller, or drive unit malfunctions that halt overall system operations. These discrete state definitions are informed by expert knowledge, maintenance records, and diagnostic standards commonly used in industrial practice, ensuring that each state has a clear physical interpretation and can be inferred through condition monitoring data or manual inspection.

The established topology, combined with the CPT, constitutes the BN of the CNC system, forming the foundation for constructing its DBN. In this illustrative example, the one-step TM is defined with a two-month interval. If the objective of this study is to monitor the reliability variations of CNC machines over the first two years following the commencement of production, the DBN will comprise 12 time slices. The network structure at each time slice remains consistent with the configuration depicted in Fig. 10. Furthermore, within each time slice, component nodes are interconnected through directed edges extending from the previous time step to the subsequent one, effectively capturing the temporal evolution of system reliability.

6.3. Dynamic reliability assessment of aluminum alloy wheels production line

Building on the constructed DBN model of the CNC machine and applying the MSS reliability dynamic updating method outlined in Section 3, the state probability distribution of the CNC machine can be updated under the multi-state model assumption. By utilizing the design parameters and historical reliability data of each manufacturing unit, the one-step state TM for each manufacturing unit is obtained as follows.

(1) The one-step TM of components in m_1

c_1: [0.999 0.001;0 1].
c_2: [0.998 0.002;0 1].
c_3: [0.992 0.004 0.004; 0 0.955 0.045; 0 0 1].
c_4: [0.987 0.013;0 1].
c_5: [0.995 0.003 0.002; 0 0.989 0.011; 0 0 1].
c_6: [0.997 0.002 0.001; 0 0.982 0.008; 0 0 1].
c_7: [0.995 0.005;0 1].
c_8: [0.99 0.005 0.005; 0 0.985 0.015; 0 0 1].
c_9: [0.980 0.015 0.005; 0 0.986 0.014; 0 0 1].
c_{10}: [0.998 0.002; 0 1].

(2) The one-step TM of components in m_2

c_1: [0.999 0.001;0 1].
c_2: [0.998 0.002;0 1].
c_3: [0.997 0.002 0.001; 0.00 0.965 0.035; 0 0 1].
c_4: [0.952 0.048;0 1].
c_5: [0.996 0.002 0.002; 0 0.993 0.007; 0 0 1].
c_6: [0.997 0.002 0.001; 0 0.982 0.018; 0 0 1].
c_7: [0.997 0.003;0 1].
c_8: [0.993 0.004 0.003; 0 0.99 0.01; 0 0 1].
c_9: [0.986 0.012 0.002; 0 0.991 0.009; 0 0 1].
c_{10}: [0.998 0.002; 0 1].

(3) The one-step TM of components in m_3

c_1: [0.999 0.001;0 1].
c_2: [0.998 0.002;0 1].
c_3: [0.997 0.002 0.001; 0.00 0.965 0.035; 0 0 1].
c_4: [0.952 0.048;0 1].
c_5: [0.996 0.002 0.002; 0 0.993 0.007; 0 0 1].
c_6: [0.997 0.002 0.001; 0 0.982 0.008; 0 0 1].
c_7: [0.997 0.003;0 1].
c_8: [0.993 0.004 0.003; 0 0.99 0.01; 0 0 1].
c_9: [0.986 0.012 0.002; 0 0.991 0.009; 0 0 1].
c_{10}: [0.998 0.002; 0 1].

Assuming that all components are initially in their best state and using the previously defined one-step transition probability matrix as the initial parameter, the state probability distribution of the CNC machine system can be inferred. To evaluate the reliability of the production line six months after commissioning, the state probability distributions for each CNC machine at the fourth time slice were extracted. By applying the prior classification of CNC machine states and defining the capacities associated with each state, the data required to assess the production line's reliability are obtained, as presented in Table 2.

If the production demand at this time is 300 qualified products, and the buffer capacities of b_1 and b_2 are 400, the minimum production capacity required for each manufacturing unit can be calculated using the MSMS reliability dynamic updating method proposed in this paper.

Table 2
The production data of CNC machines.

Manufacturing unit	State	Capacity[a]	Probability
m_1 (Yield rate: 0.95)	0	400	0.8125
	1	320	0.0621
	2	150	0.0007
	3	0	0.1247
m_2 (Yield rate: 0.96)	0	350	0.7676
	1	280	0.0466
	2	90	0.0004
	3	0	0.1854
m_3 (Yield rate: 0.98)	0	380	0.7860
	1	300	0.0264
	2	160	0.0071
	3	80	0.0002
	4	0	0.1803

Notes: [a]Units of raw materials/WIP that each manufacturing unit can process per unit time.

The specific calculation is as follows.

First, the required amount of input raw materials is calculated based on the target output quantity. Since the rework of m_3 is considered and limited to only one rework cycle, the production process includes two feasible paths, which are $P_1^* = (m_1, m_2, m_3, b_1, b_2)$ and $P_2^* = (m_1, m_2, m_3, m_3, b_1, b_2)$. Hence, the amount of input raw materials is:

$$Input = \frac{output}{p1p2p3 + p1p2(1-p3)p3} = \frac{300}{0.95 \times 0.96 \times 0.98 + 0.95 \times 0.96 \times (1-0.98) \times 0.98} = 329$$

The MP represents the minimal capacity vector required to ensure that the production line can generate enough products to meet the task demand. It is determined based on the input materials and the required output. Buffer capacities are sufficiently large and do not cause congestion in the production process, so their impact on reliability assessment can be ignored in this example. As rework for m_3 is considered, the MSMS contains two MPs. The product flow for the two feasible paths is as follows:

$$P_1^* : (329 \times 0.95, 329 \times 0.95 \times 0.96, 329 \times 0.95 \times 0.96 \times 0.98) = (312.55, 300.05, 294.05)$$

$$P_2^* : (300 \times 0.95, 300 \times 0.95 \times 0.96, 300 \times 0.95 \times 0.96 \times (1-0.98) \times 0.98) = (312.55, 300.05, 299.92)$$

The MP of the aluminum alloy wheel production line is:

$$MP = X^{\min} = (x_1^{\min}, x_2^{\min}, x_3^{\min}) = (320, 350, 300)$$

According to the state definition, the reliability of the aluminum alloy wheel production line in this example is calculated as follows:

$$R = 1 - (1 - r_1^*)(1 - r_2^*) \cdots (1 - r_i^*)$$
$$= 1 - (1 - 0.8125 \times 0.7676 \times 0.7860)(1 - 0.0621 \times 0.7676 \times 0.7860)(1 - 0.8125 \times 0.7676 \times 0.0264)(1 - 0.0621 \times 0.7676 \times 0.0264)$$
$$= 1 - 0.5098 \times 0.9625 \times 0.9835 \times 0.9987 = 0.5180$$

The calculated reliability indicates that six months after commissioning, the aluminum alloy wheel production line has a probability of 0.5453 of successfully meeting the production target of delivering 300 qualified products.

At this stage, the reliability data is derived solely from historical statistics. To achieve dynamic reliability updating, observational data is incorporated at the second and third time slices. At the second time slice, namely two months after commissioning, the system-level observational information pertaining to m_2 is obtained and utilized to update the system reliability. Additionally, the observation collected from the c_3 of m_2 at the third time slice is utilized to update the system reliability as well. Due to the epistemic uncertainty in the observations, the resulting data is represented as evidential observational information, with the corresponding contour functions given as $(1, 0.08, 0.08, 0.08)$ and $(1, 0.3, 0.3)$. By applying the constructed DBN model and integrating the evidential observational data, the updated state probability distributions for m_2 are obtained, as presented in Table 3.

After updating the state probability distributions based on real-time observations, the reliability of the aluminum alloy wheel production line six months after commissioning, under the demand $d = 300$, is recalculated as:

$$R_{update} = 1 - (1 - 0.8125 \times 0.8405 \times 0.7860)(1 - 0.0621 \times 0.8405 \times 0.7860)(1 - 0.8125 \times 0.8405 \times 0.0264)(1 - 0.0621 \times 0.8405 \times 0.0264)$$
$$= 1 - 0.4632 \times 0.9589 \times 0.9820 \times 0.9986 = 0.5644$$

The updated results indicate that six months after commissioning, the actual degradation of the aluminum alloy wheel production line is slower than anticipated. With a task demand of producing 300 qualified products, the probability of task completion is 0.5972.

To validate the accuracy of the results, the reliability of m_2 is further examined throughout the updating process. The system is assumed to be reliable when m_2 is in states 0 and 1, and the corresponding updated reliability function is illustrated in Fig. 11. In this figure, the blue curves indicate the reliability function of m_2 derived from historical data. After incorporating observational data at the second and third time slices, the revised reliability function is obtained and shown in red. This comparison highlights the influence of real-time observational updates and demonstrates the improvement in reliability assessment achieved through the dynamic reliability updating method.

Fig. 11(b) presents a locally magnified view of Fig. 11(a), offering a more detailed perspective on the reliability update process. As observed in Fig. 11, the actual reliability of m_2, obtained after integrating the two observational updates, is higher than the initial predicted reliability. This result aligns with the previous reliability updates for the aluminum alloy wheel production line, further confirming the effectiveness and consistency of the dynamic reliability updating approach.

To illustrate the advantages of the proposed method, a comparative analysis was conducted against a baseline MFN-based model. Traditional MFN approaches for MSMS reliability evaluation primarily determine system reliability based on predefined state probability distributions of individual components or manufacturing units. These models generally depend on assumptions or fixed probability distributions derived from historical data, without incorporating the real-time operational variability that frequently occurs during actual system operation. For example, in References [6–8], the state probability distribution and productivity of each manufacturing unit are directly provided as machine data, while in Reference [20], the state probability distribution is derived by specifying the state transition intensity matrix of the machine. These approaches generally neglect the influence that variations in unit-level state probabilities may exert on the overall MSMS reliability. From a theoretical perspective, if the state probability distributions of manufacturing units are determined solely through expert judgment, the resulting accuracy tends to be lower compared to that obtained from parameterized models, due to the inherent subjectivity associated with expert assessments. In contrast, incorporating real-time observational data for dynamic updating introduces more objective and informative evidence, which makes the resulting distributions more representative of the actual operating conditions. In this article, the proposed model extends existing reliability assessment frameworks by incorporating imprecise observational data of component and subsystem states into a dynamic inference process. Through the use of DBNs, the system captures temporal dependencies and updates the reliability distribution of each node based on evidential observations. This enables a more accurate reflection of component degradation and supports inference of system-level behavior, offering a theoretically grounded method to handle both aleatory and epistemic uncertainties. Practically, the model provides valuable decision support in the maintenance and operation of MSMS. The dynamic evaluation of task reliability allows engineers to prioritize components for inspection and maintenance based on real-time evidence, rather than relying solely on historical averages. This facilitates predictive maintenance planning, improves resource allocation, and reduces unexpected downtime. For instance, in the illustrative case, if the reliability of aluminum alloy

Table 3
The updated production data of m_2.

Without observation		After the first updating		After the second updating	
State	Probability	State	Probability	State	Probability
0	0.7676	0	0.8380	0	0.8405
1	0.0466	1	0.0338	1	0.0339
2	0.0004	2	0.0002	2	0.0002
3	0.1276	3	0.1280	3	0.1254

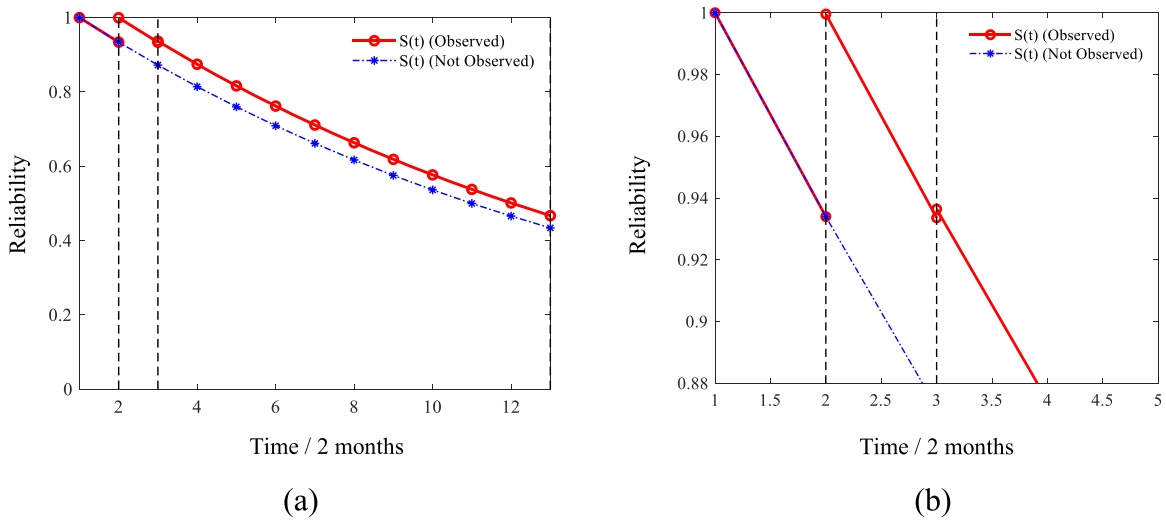


Fig. 11. The reliability function of m_2.

wheels production line cannot meet the required task, decision-makers can develop proactive strategies to mitigate risks to prevent failures that may hinder the production of sufficient product.

The proposed DBN framework assumes a fixed two-month time slice and employs first-order Markovian transitions to model the evolution of manufacturing unit states. These simplifications are made for practical reasons. In many industrial environments, maintenance planning and inspection activities are conducted on monthly or bi-monthly cycles, making the chosen granularity suitable for supporting real-world decision-making. The first-order Markov assumption, wherein the current state is influenced solely by the immediately preceding state, is commonly employed in probabilistic modeling due to its analytical simplicity and effectiveness in capturing short-term dependencies. However, such assumptions may not adequately represent systems in which state transitions possess memory effects extending beyond a single time step or occur across irregular temporal intervals. Consequently, evaluating the model's performance and sensitivity to these temporal assumptions constitutes an important area for future research. Potential enhancements include the adoption of variable-length time slices, higher-order Markov models, or the implementation of semi-Markov and hierarchical temporal Bayesian models, which offer improved capacity to represent complex system dynamics while preserving computational tractability.

7. Conclusion and future work

This paper proposes a method for updating the reliability of MSMSs by merging multi-level imprecise observations of machines. The epistemic uncertainty associated with the observations is quantified using mass and contour functions within evidence theory. A DBN is employed to update the system state distribution and reliability based on these multi-level imprecise observations. To demonstrate the effectiveness of the proposed approach, an aluminum alloy wheel production line is used as an illustrative example. In this case study, the integration of observational information changes the state probability distributions of machines, leading to the reliability updating for the production line. Since the machine is observed to be in an optimal state, it can be inferred that the degradation of aluminum alloy wheel production line is progressing at a slower rate than initially predicted. The updated reliability function aligns with the results of the system reliability updating, further validating the effectiveness of the proposed dynamic reliability updating approach.

While the proposed method improves the accuracy and adaptability of reliability assessment, the integration of DBNs and evidence theory inevitably introduces considerable computational complexity. This complexity arises from the structure of the DBNs, the temporal depth of analysis, and the fusion of imprecise observational information, which may constrain the scalability of the approach in large-scale or real-time applications. Future research will prioritize enhancing computational efficiency, investigating alternative methods for uncertainty quantification, and extending the framework to support lifecycle prediction and maintenance optimization. These advancements aim to strengthen reliability management in complex manufacturing environments. Furthermore, subsequent efforts will focus on refining the proposed methodology by exploring novel techniques for quantifying epistemic uncertainty and improving system state derivation. In addition, after determining system reliability, future studies will aim to forecast the MSMS lifecycle and support optimal maintenance decision-making, thereby contributing to more effective and informed reliability strategies in advanced manufacturing systems.

CRediT authorship contribution statement

Tudi Huang: Writing – original draft, Software, Data curation, Conceptualization. **Qin Zhang:** Writing – review & editing, Validation. **Michael Beer:** Writing – review & editing. **Yu Liu:** Writing – review & editing, Funding acquisition. **Hong-Zhong Huang:** Writing – review & editing, Visualization, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

A.1 CPT of the Subsystem 1 in the CNC Machines

	$C_1(t)$																							
	0												1											
$C_2(t)$	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1
$C_3(t)$	0	0	1	1	2	2	0	0	1	1	2	2	0	0	1	1	2	2	0	0	1	1	2	2
$C_4(t)$	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
$S_1(t)$	0	2	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2

A.2 CPT of the Subsystem 2 in the CNC Machines

	$C_5(t)$																	
	0						1						2					
$C_6(t)$	0	0	1	1	2	2	0	0	1	1	2	2	0	0	1	1	2	2
$C_7(t)$	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
$S_2(t)$	0	2	1	2	2	2	1	2	1	2	2	2	2	2	2	2	2	2

A.3 CPT of the Subsystem 3 in the CNC Machines

	$C_8(t)$																	
	0						1						2					
$C_9(t)$	0	0	1	1	2	2	0	0	1	1	2	2	0	0	1	1	2	2
$C_{10}(t)$	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
$S_3(t)$	0	2	1	2	2	2	1	2	1	2	2	2	2	2	2	2	2	2

A.4 CPT of the whole system of the CNC Machines

	$S_3(t)$								
	0			1			2		
$S_4(t)$	0	1	2	0	1	2	0	1	2
$S(t)$	0	1	3	1	2	3	3	3	3

Data availability

No data was used for the research described in the article.

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